

VERNAL, UT, USA

WMO: 725705

Lat: 40.443N

Lon: 109.513W

Elev: 5270

StdP: 12.10

Time Zone: -7.00 (NAM)

Period: 98-19

WBAN: 94030

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.5	(c) -0.2	(d) -10.5	(e) 3.8	(f) -4.0	(g) -5.9	(h) 4.9	(i) 2.7	(j) 17.0	(k) 28.1	(l) 12.5	(m) 28.5	(n) 1.8	(o) 280	(p) 0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 30.4	(c) 93.4	(d) 59.8	(e) 91.1	(f) 59.3	(g) 88.8	(h) 58.8	(i) 64.0	(j) 82.9	(k) 62.5	(l) 81.8	(m) 61.4	(n) 81.3	(o) 7.6	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.4	88.9	68.3	56.2	81.9	68.7	54.4	76.6	68.9	31.7	81.9	30.7	81.3	29.8	81.0	69.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.8	17.5	14.2	-8.0	97.1	6.5	1.8	-12.7	98.3	-16.5	99.4	-20.1	100.4	-24.8	101.7		
			-8.6	65.9	6.4	1.6	-13.2	67.0	-16.9	67.9	-20.5	68.8	-25.1	69.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	18.8	26.9	39.7	47.6	56.9	67.3	74.6	71.1	62.1	48.4	35.2	21.3
	DBStd	20.02	10.05	9.38	7.55	6.87	7.46	6.83	3.87	3.97	6.43	7.34	7.57	9.68
	HDD50	3544	966	646	327	123	25	1	0	0	5	116	446	889
	HDD65	7017	1431	1066	784	522	264	55	0	6	124	516	895	1354
	CDD50	2668	0	0	9	51	238	520	763	653	368	65	1	0
	CDD65	667	0	0	0	0	13	124	299	194	37	0	0	0
	CDH74	9456	0	0	1	18	394	2080	3829	2396	718	20	0	0
	CDH80	4052	0	0	0	2	88	883	1897	988	193	1	0	0
Wind	WSAvg	5.0	2.7	3.7	5.5	7.0	6.6	6.5	5.5	5.3	5.1	4.7	4.0	3.0
Precipitation	PrecAvg	8.5	0.5	0.5	0.7	0.8	0.9	0.7	0.6	0.7	0.9	1.2	0.6	0.6
	PrecMax	12.1	2.0	1.3	2.2	2.2	2.8	2.3	2.0	2.4	2.9	4.4	1.5	1.9
	PrecMin	3.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.2	0.4	0.4	0.6	0.5	0.7	0.7	0.5	0.5	0.7	1.0	0.4	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.5	54.7	70.3	76.3	87.6	95.0	97.4	93.9	89.0	76.9	61.2	47.8
		MCWB	36.5	40.2	47.3	49.9	57.5	58.9	61.9	59.9	58.0	52.4	44.5	38.5
	2%	DB	39.4	49.4	64.2	72.0	82.0	91.5	94.7	91.1	84.9	71.7	56.6	42.1
		MCWB	32.7	37.9	44.4	47.9	54.4	57.4	60.6	59.5	56.6	51.0	42.8	34.3
	5%	DB	36.1	44.7	60.3	68.2	78.8	88.9	92.4	88.8	81.7	67.1	52.9	37.6
		MCWB	31.3	35.7	42.6	46.3	52.7	56.9	60.2	59.1	56.1	49.4	40.9	31.7
	10%	DB	32.8	39.9	55.1	64.0	74.6	85.4	90.1	86.1	78.6	63.2	48.3	34.3
		MCWB	29.5	33.3	40.4	45.0	51.2	56.0	59.9	58.7	54.6	47.0	38.8	29.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.8	41.8	48.3	51.6	59.1	61.4	66.2	65.5	62.3	55.4	46.7	39.2
		MCDB	42.7	52.4	67.1	73.6	83.6	86.6	85.4	84.1	79.1	71.4	57.9	46.3
	2%	WB	34.4	38.7	45.7	49.2	55.8	59.9	64.5	63.6	60.0	52.9	44.1	35.4
		MCDB	38.0	46.8	62.6	66.9	76.7	84.7	83.9	81.1	76.7	66.1	53.5	40.0
	5%	WB	32.0	36.5	43.4	47.6	54.1	58.7	63.2	62.2	58.4	50.9	41.9	32.7
		MCDB	35.2	43.3	57.8	64.8	74.3	82.9	82.9	80.0	74.4	64.0	51.0	37.0
	10%	WB	29.6	34.1	41.3	45.9	52.4	57.3	62.1	61.1	56.8	48.8	39.5	30.3
		MCDB	32.8	39.4	54.4	62.2	71.7	80.7	82.6	79.0	72.2	60.9	47.5	34.0
Mean Daily Temperature Range	5% DB	MDBR	18.2	19.6	24.4	26.1	27.7	30.9	30.4	29.7	28.8	25.4	21.6	17.4
		MCDBR	20.5	23.8	32.1	33.5	34.1	34.9	33.5	33.8	33.6	30.4	27.1	21.7
		MCWBR	14.7	14.8	16.3	15.3	13.6	12.6	11.5	12.0	13.3	14.7	15.7	15.1
	5% WB	MCDBR	18.1	21.3	29.8	29.8	31.0	31.7	28.3	27.7	27.4	27.1	24.7	19.3
MCWBR		13.5	13.8	15.5	14.2	12.9	12.2	10.1	10.6	11.5	13.3	14.8	14.0	
Clear-Sky Solar Irradiance	taub	0.254	0.265	0.282	0.305	0.317	0.318	0.354	0.357	0.310	0.288	0.262	0.251	
	taud	2.523	2.530	2.511	2.456	2.477	2.490	2.428	2.417	2.531	2.562	2.573	2.555	
	Ebn at Noon	296	307	311	308	304	302	290	287	297	294	291	289	
	Edh at Noon	23	26	30	34	34	34	36	35	29	25	22	20	
All-Sky Solar Radiation	RadAvg	693	989	1428	1819	2125	2451	2238	1996	1713	1227	831	606	
	RadStd	62	81	98	101	136	154	107	84	107	84	64	58	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page